

Toy Design . Graduation Project

INDUSTRY INTERNSHIP

Designing STEAM DIY Toys

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Woxsen University

Who I am , and what this is

For my graduation project, I spent five months as a Toy design intern at OIMake. This is the story of what I observed in the field, how kids interact with toys and how they perceive them.

Designer

D Sri Charan Reddy

Internship

OiMake · Jan–May 2026

Programme

Graduation Project · Woxsen University

Duration and focus

2 products and 1 POC across 5 months

INTERNSHIP CONTEXT

A Full Product Pipeline

This was not just an observation placement. Over 5 months, I was embedded within OiMake's product team—working across research, design, prototyping, testing, and production.

Commercially Live

Products designed here are not academic concepts — they are live on the market.

Segment

DIY STEAM Toys — hands-on educational kits for children aged 8+.

ABOUT OIMAKE

Born from One Child's Curiosity

OiMake (formerly MakerVerse) was founded when a 9-year-old wanted to understand how electronics work—by doing, not watching. It started with light-up birthday cards and Bluetooth circuits and evolved into a full DIY kit brand.



Everything in One Box

Tools, components, and visual manuals — complete and self-contained.



Made in India

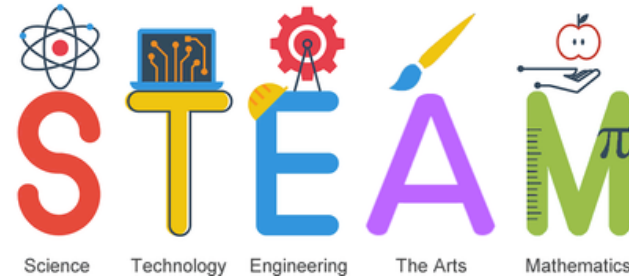
Mission: cultivate a new generation of makers.



Where Learning Meets Making

What is STEAM?

Science · Technology · Engineering · Art · Mathematics —
real materials and real processes in children's hands.
Learning happens **through making**, not through instruction.



The Market Opportunity

India's edutoy market is rapidly growing yet massively
at the ₹500–1,200 price point. Most toys on platforms are
sold. That's where we concentrate.

Toys priced above ₹2,000, creating a clear **accessibility gap**.

Our play: Affordable + integrated + hands-on STEAM
within reach.

THE PROBLEM OIMAKE SOLVES

The Gap Between Passive Play and Real Learning

Construction Kits

Plateau quickly—no re-engagement after the first build.

Coding Kits

Steep learning curve — most abandoned before completion.

Robotic Kits

A child interacts with a finished product — builds nothing, learns less.

- ✓ **OiMake's Answer:** Kits where the child builds something real, keeps it, and uses it daily — a wearable, functional, personal object. Not a science experiment on a shelf.

TARGET USERS

The Child Uses It. The Parent Buys It.

The real customer is not the child—it is the parent. Parents drive 100% of the purchase decision in this segment. The product must communicate learning value instantly at the point of purchase.

- 📌 Parents are not buying play — they are buying **preparation**. A product that credibly positions itself as screen-free and skill-building wins immediately.

Two Truths That Define This Market

Truth 1 – Anxiety

Parents are genuinely worried about what screens are doing to their children. They feel guilt when their child is on a phone for hours. A hands-on kit **relieves that guilt** — it feels like good parenting.

Truth 2 – Aspiration

Urban, educated Indian parents want children who build, create, and problem-solve. A kit that teaches electronics and art signals forward-thinking parenting. It is a purchase driven by identity as much as function.



How Parents Evaluate Before They Buy

Factor	What Parents Need to See
Engagement Duration	45–90 min of focused, screen-free activity
Learning Proof	Visible, explainable outcomes — circuits, motor skills, creativity
Screen Time Anxiety	A credible alternative to phone time
Age Appropriateness	Clear age label = purchase confidence
Value for Money	₹500–1,200 sweet spot — enough to trust, affordable enough to try
Type of Activity	Build-it-yourself strongly preferred over passive play

KIDS IN WORKSHOPS

Touch First.

Children physically explore any unfamiliar object before consulting instructions. —always. First response: touch, rotate, attempt to disassemble. A toy that rewards this instinct builds trust before any manual is opened.

Social Play Emerges

When two children had the same kit, activity shifted to comparison and negotiation — "Why does mine spin faster?" "Mine glows brighter."





PROJECT 1

LED Light Bag — Paint It. Build It. Wear It.

A DIY STEAM kit for children aged 8+. The child paints a schematic illustration on a cotton drawstring bag, assembles a 5-LED PCB from the kit, and slides it into a pocket behind the painted surface. Flip the switch — their artwork glows.

45–60

No adult required

₹893

Live on website

~100kits

Already in children's hands



Secondary Research: Market Landscape

What already exists. And what's missing.

Brand	What They Offer	The Gap
Smartivity	STEAM / Mechanical kits	No art, no textile
Adafruit / SparkFun	Art + Electronics	Requires coding & soldering — not child-friendly
Generic Colouring Bags (India)	Painting only	Zero electronics
Lumen Couture / TechnoChic	LED wearables	No painting, adult-targeted

✅ **Market gap confirmed:** No product under ₹ 1,000 in India combines fabric art + guided LED assembly. LED LightBag fills it at ₹893.

sl. No	link	product and price	recent sales	images
1	https://amazon.in/Bloomingo-DIY-Kitty-Bag-Kit-for-Girls/	₹444 Bloomingo DIY Kitty Bag Kit for Girls/	50+ last month	 
2	https://www.amazon.in/Arishto-DIY-Bunny-Crochet-Bag-Kit-for-Kids/	₹999 Arishto DIY Bunny Crochet Bag Kit for Kids	no info	 
3	https://www.amazon.in/GIGGL-DIY-Fancy-Kids-Sewing-Bag-Kit-Sew-Your-Own-Felt-Tote-Creative-Craft-Activity-for-Boys-Girls/	₹299 Sewing Craft Kit for Kids Handmade Educational Project Random Colour Pack 1	no info	 
4	https://www.amazon.in/-/hi/PO-PokuPop-DIY-Fancy-Kids-Sewing-Bag-Kit-Sew-Your-Own-Felt-Tote-Creative-Craft-Activity-for-Boys-Girls/	₹199 PokuPop DIY Fancy Kids Sewing Bag Kit – Sew Your Own Felt Tote, Creative Craft Activity for Boys & Girls	no info	 
5	https://www.amazon.in/HASTHIP-DIY-Handmade-Bag-Material-Kit-Girls-Purse-Bag-Hardware-Chain-Design-Shoulder-Bag-Material-Kit-DIY-Lace-Weaving-Bag-Material-Kit-with-Tools-Accessories-DIY-Bag-Gift-for-Girls/	₹561 HASTHIP® DIY Handmade Bag Material Kit Girls Purse Bag Hardware Chain Design Shoulder Bag Material Kit DIY Lace Weaving Bag Material Kit with Tools & Accessories DIY Bag Gift for Girls	4.2* (53 global	
		₹668 HASTHIP® Plastic Mesh		

sl. No.	form of the bag	platform and link	product detail	material	images
1	sling bag	deo dap -- https://deodap.in/products/mermaid-tail-shaped-kids-sling-bag-1-pc?variant=53042824806710&country=IN&currency=INR&utm_medium=product_sync&utm_source=google&utm_content=sag_organic&utm_campaign=sag_organic&srsltid=AfmBOocMFnVG-KBRetlJFdOw-DBMn7eb3Sfw8wngjZqmxkZIVd1MylQE0	29rs Mermaid Tail Sling Bag	Fabric with Sequins & Synthetic Tail Design	 
2	cross body bag	ARHAMSMART - The Wholesale Mart https://arhamsmart.com/product/kids-crossbody-sling-chest-bag/?attribute_pa_kidschestbag=rainbow-unicorn&srsltid=AfmBOoq_gtDyYiD3iWdUWIZSjuSk8j8ro9QFgjqX0Cqj6F9KFBDxiGVMOfg	₹110 Kids Crossbody Sling Chest Bag		
3	totebag	India mart - https://www.indiamart.com/proddetail/canvas-tote-bag-22889698355.html?utm_medium=prd_ads&utm_campaign=23411315669&utm_ad_group=194444872761&utm_content=product&utm_source=google&gclid=Cj0KCQiApfjKBhC0ARIsAMiR_IsB-wdAAmccS1Qc2A7Mk3RAI_iQCcyUBhF9_5XhPXR4qsq8sqQc1eEaAs0QEALw_wcB	Plain Cotton Canvas Tote Bag, upto 7 kg		

Form Factor & Ergonomics

Why a drawstring bag?

Formats Explored

Backpack · Tote · Sling · Crossbody · Drawstring ✓

Why Drawstring Won

Compact — proportioned for ages 8–14

Large flat front panel — ideal painting canvas & lighting surface

Timeless, familiar format — no learning curve



Anthropometric Data (Ages 8-14)

Bag Width

Shoulder width 28-39cm →
30cm

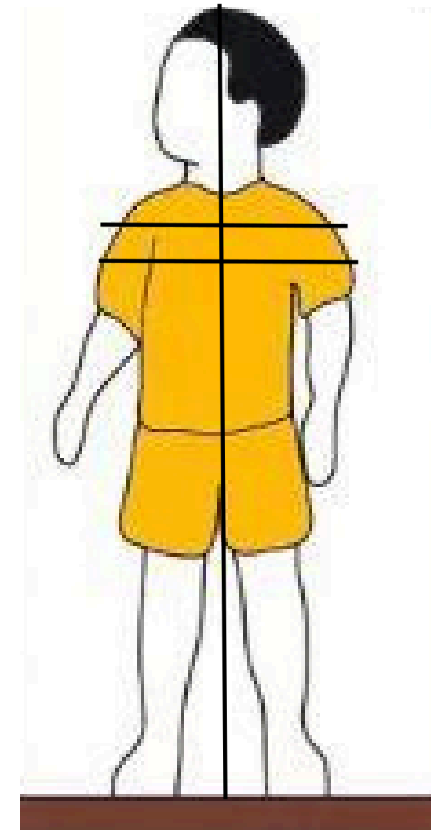
Back Length 30 -46 cm
→ 40cm

Bag Height

Back Dimensions in
cm → 30 cm in width
40cm in Height

Motor Skills

Fine motor development sufficient for LED assembly
at age 8+



Material & Painting Exploration

Finding the right surface. Finding the right tool.

Materials Tested

Non-woven · PropyleneCanvas ·
Wool Felt · Synthetic · Muslin ·
Bleached Muslin

✓ Winner: Muslin — paints vibrantly, soft texture, safe, durable, low cost

Mediums Tested

Acrylic paints · Acrylic markers ·
Fabric markers · Sketch pens ·
Washable pens

✓ Winner: Fabric markers — controlled, minimal mess, non-toxic, child-safe



Tech Integration & Lighting Evaluation

Making LEDs work beautifully inside fabric.

LED Types Compared

DIP · SMD · COB — evaluated on brightness, size, flexibility, and integration ease.



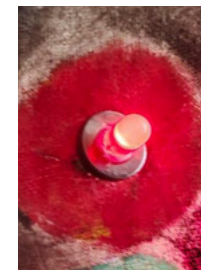
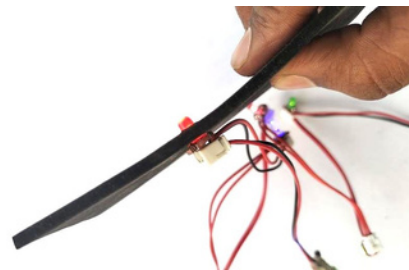
Integration Methods Explored

Direct threading — not durable, unsafe

Adhesive mounting — too complex for children

Button Over the Pocket — The whole led is exposed

Custom PCB + EVA foam + Bag Pocket ✓



The final system is safe, durable, child-assembled, and removable for washing.

Illustration Development



Market Observation

Studied toy stores, children's books, cartoon merchandise, and DIY kits to identify what illustration styles children respond to.



Key Findings

Bold, simple visuals win. Top themes: animals, fantasy, cute characters, nature. Thick outlines, bright colours, minimal complexity.



Design Output

LED positions are deliberately aligned with key illustration elements — graphics function as a creative guide + lighting map simultaneously.





Vehicle



Places



Animals



Objects



Emotions



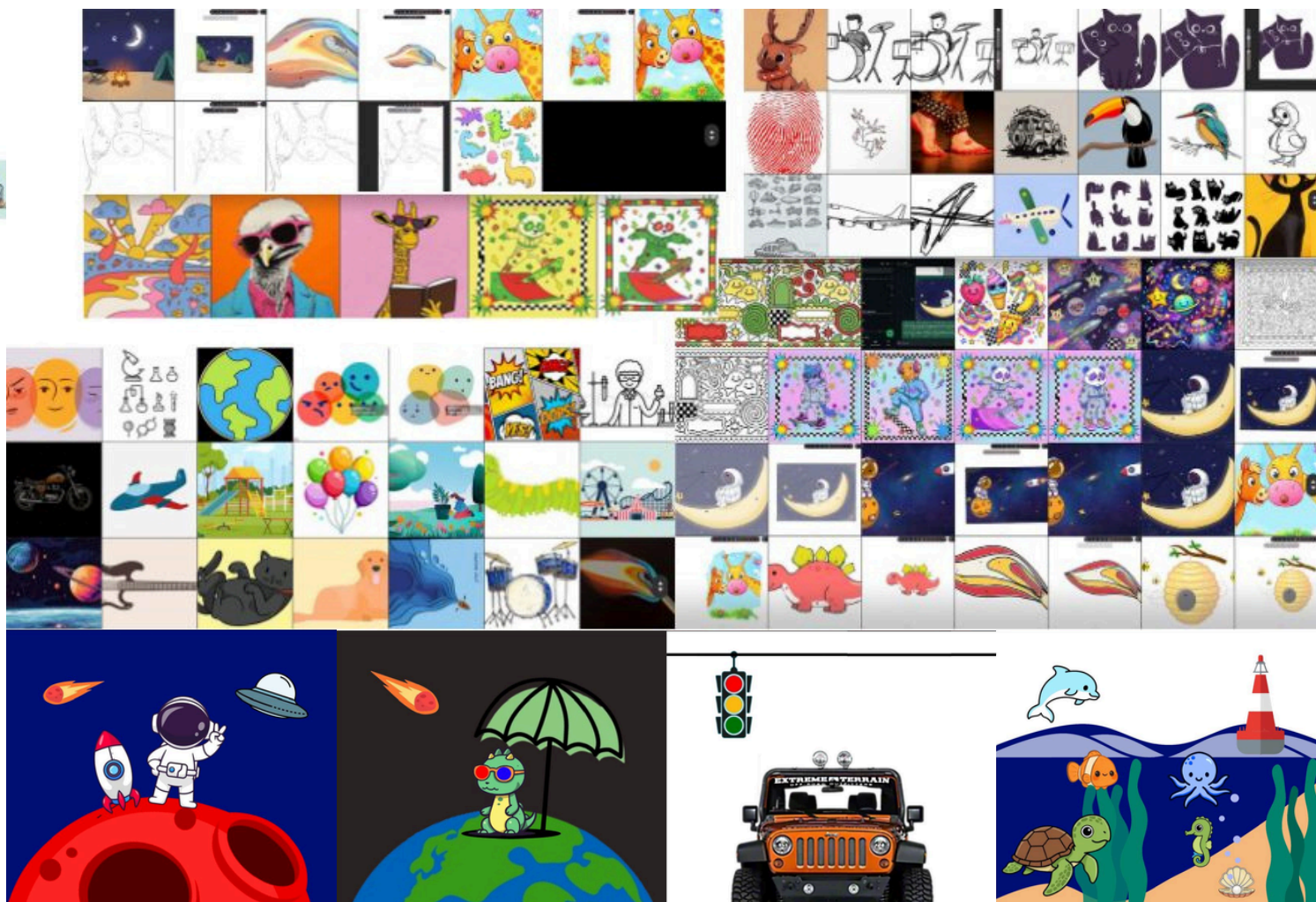
Comic text



Tasks



Missilinous



Prototyping

From sketch to something a child can hold.

What Was Tested

- Lightdistribution · battery performance · glow quality · assembly ease
- LED placement must align precisely with illustration
- Even distribution critical — harsh or dim pockets break visual impact
- EVA foam thickness iterated for optimal diffusion
- Component fit tested for correct child-hand scale



User Testing & Feedback

What We Observed

Hands-on workshops were conducted with children to test the usability and engagement of the DIY LED backpack kit. Using a photo-based instruction manual, children independently painted, assembled, and completed the DIY activity with minimal guidance.



Changes Made

→ Simplified outlines

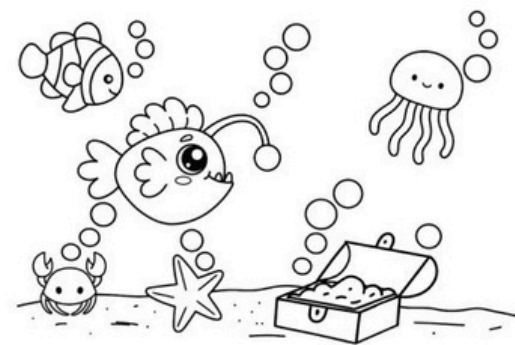
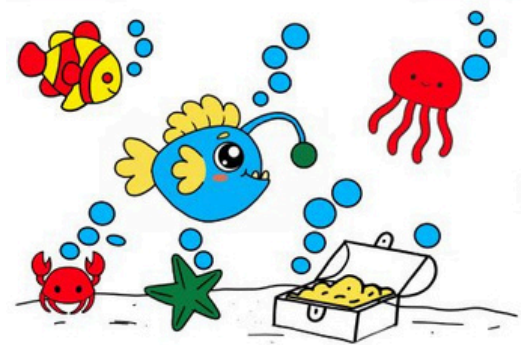
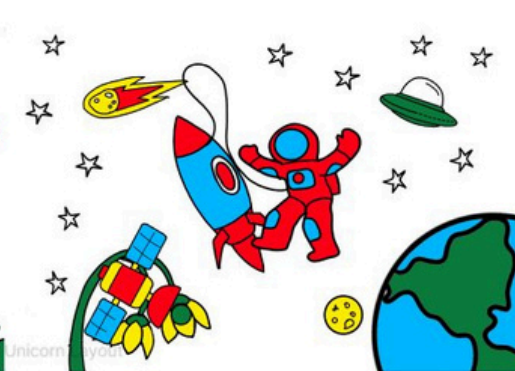
Easier tracing for younger children

→ Reduced colour count

Removed visual clutter

→ Easy Assembly

Made the Eva and the Bag holes bigger



Instruction Manual

Design Principles

Real process photos converted to simplified illustrations — no text instructions

Sequential layout: each step shows hand position + component

Includes: component overview · assembly guide · colouring instructions · safety notes

✓ **Result:** All children completed independently. Zero adult assistance required in final testing.



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LED Light Bag

DIY Guide

Version 1.1



 Oimake

1

Bill of Materials

Every component. Every cost. Accounted for.

Key Components

Bag & Surface

Muslin drawstringbag · Pre-printed illustration

Art Tools

Fabricmarkers (set)

Electronics

Custom 5-LEDPCB · EVA foam diffuser · Semi-transparent pocket · AA battery holder · On/off switch · Wires

Packaging

Visualinstruction manual · Branded sleeve packaging

Pricing Logic

BOM cost + Logistics + packaging + Marketing + margin

₹1099

₹500–1,200

Parent sweet spot — trust + affordability

 Positioned below all comparable products.

Packaging: Sleeve Design

Standard box. Branded sleeve. Assembled in-house.

Cost Efficiency

Standard pre-ordered box with custom branded sleeve printed externally — keeps unit cost low.

In-House Assembly

Sleeving and final kit assembly done in-house — reduces production overhead.

At-a-Glance Communication

Sleeves clearly convey age group, activity type, and learning outcomes — instantly.





Free shipping on all orders



Shop

Summer Camps

Make @ Home Program

About

Blog



Makerverse is now OIMake. Same DIY STEM kits, same smiles, new name.



MAKERVERSE

LED Light Bag | DIY STEM Kit | 8-14 yrs | DIY Wearable Tech Kit

Rs. 1,099.00 ~~Rs. 893.00~~ Sale

Free Shipping PAN India | ISI Certified | Made in India

Quantity

- 1 +

Add to cart

Buy it now

🕒 Order today within 23:35:54, you'll receive your package between 24/05/2026 to 26/05/2026



 **Purchased**
May 20

 **Processing**
May 21 - 23

 **Delivered**
May 24 - 26

Design it, Light it up & Watch it Glow!





Project 2: Light-Up Fridge Magnet

From a client request to a return-gift product.

Origin

Client requirement — a fridge magnet with tech integration

Pivot

Repositioned as a low-cost return gift for children's birthday parties

Target Market

Birthday parties, school events, festive occasions — gifting

A glowing, child-assembled magnet = memorable, novel, skill-building gift



Form Exploration

Forms Explored

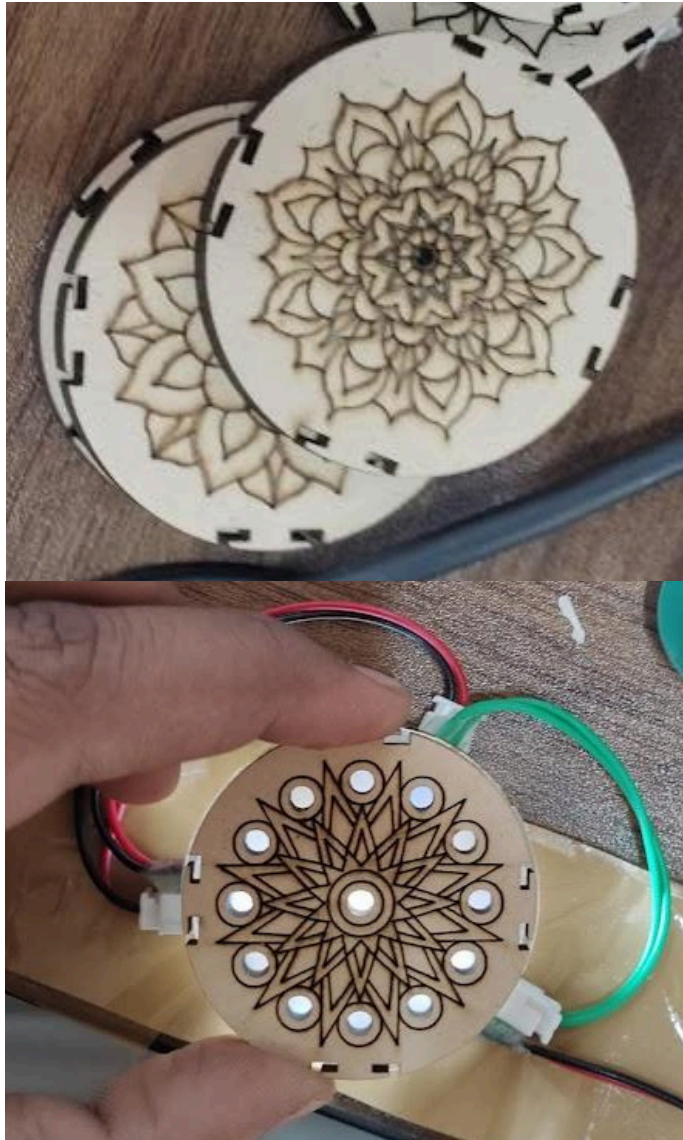
Geometric·Organic · Emblem-based

Selection Criteria

- Visual appeal for children
- Compatible with LED glow diffusion
- Easy laser cutting + MDF stacking
- Badge-like, collectible feel

✓ **Winner:** Compact circular emblem — playful, balanced, focuses attention on artwork, simplifies layer alignment.





Mandala Pattern Development

The pattern does two jobs: look good unlit, glow beautifully lit.

Evaluation Criteria

Visual density when unlit · Glow diffusion quality through cutouts when lit

Final Selection

Balanced density + visual familiarity in home décor

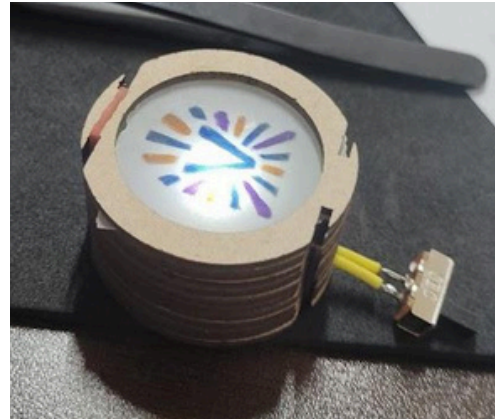
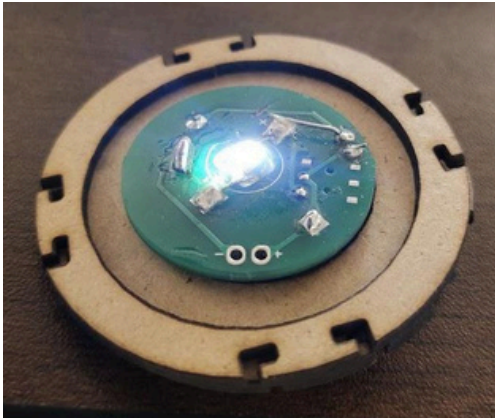
Intentional Glow

LED positions correspond to petal groups – deliberate illumination, not random scatter

Prototyping Journey: Stages 1-7

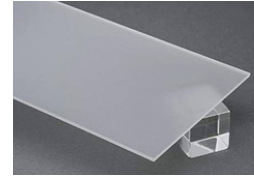
Proof of Concept

Rough MDF + single LED confirmed MDF can glow ✓ — but
single LED gave uneven, unacceptable glow



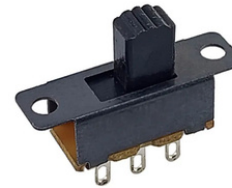
Diffusion Testing

Tracing paper, frosted acrylic, sanded surfaces tested — adjustable jig found 6–8mm gap = optimal diffusion



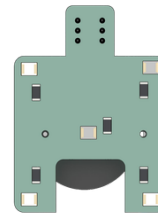
Switch Refinement

Slide switch replaced—non-intuitive, prone to accidental activation. Push-to-toggle button: clear tactile feedback ✓



LED Array

Centre + 4 angular positions → uniform glow achieved ✓



Assembly Challenge & Solution

The rubber band problem. The screw solution.

Problem

Rubber bands held MDF layers during construction — maintaining alignment while winding bands was too difficult for young children. Layers shifted, causing accuracy errors.

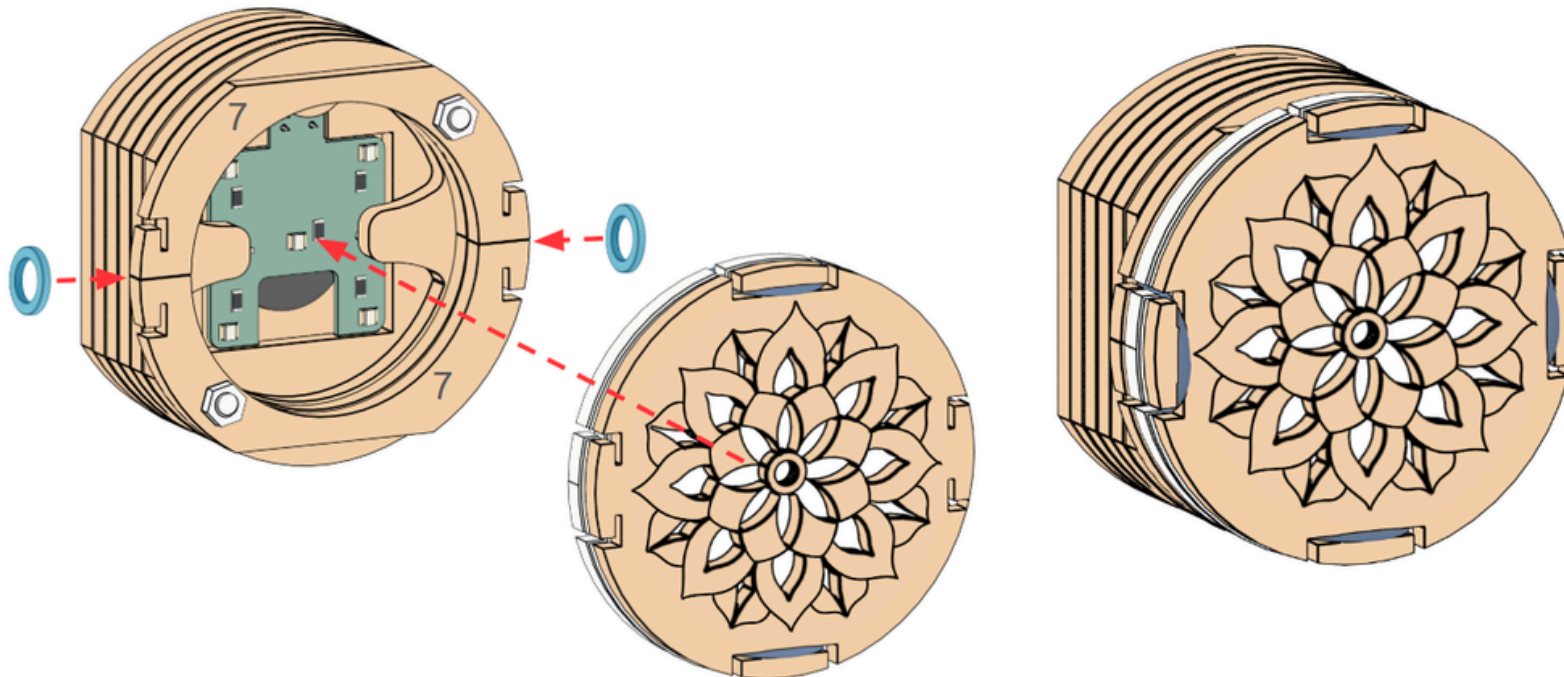


Screw-Guided Stacking System

M3 screws inserted through Layer 1 first

Layers –7 stacked directly onto screw shafts —
screws act as alignment guides

Rubber band clips applied at a defined later stage



Instruction Manual & BOM

Illustration-Based Manual

Each step shows hand position + layer placement. Tested with target age group — all completed independently. Push-button glow on completion: immediate positive response in every participant.

Key BOM Components

Laser-cut MDF layers (7)

Custom PCB · 5 LEDs

Push-to-toggle switch

CR2032 coin-cell battery

M3 screws · Rubber band clips · Magnet backing



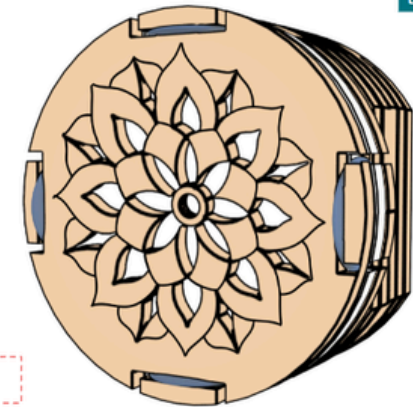
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Light up Fridge Magnet

DIY Guide

Version 1.1





Packaging: Sticker Approach

Minimal. Cost-conscious. Return-gift ready.

The Problem

Fullprinted sleeve not viable —
return gift must stay low cost

The Solution

Sticker label over standard
small box — dramatically lower
cost per unit vs. sleeve



The Result

Brand identity
maintained, suitable for bulk gifting at parties,
school events, festive occasions

Product Status & Future Launch

Production-ready. Launch incoming.



Fully Ready

Designed, tested, assembly-refined. Instruction manual complete. BOM finalised.



Planned Launch

OiMake website + Amazon
India



Market Position

India's first child-assembled, illuminated, low-cost return gift

Project 3: 5-in-1 Kit

One electronics platform. Five different toys.

Modular STEAM Kit

Same components, 5 different builds — one toy at a time, disassemble, rebuild differently

Hero Component: LDR

Light Dependent Resistor—
Light detected → Output ON ·
No light → Output OFF

Core Concept

This simple sensor logic becomes the engine of 5 completely different toys — modular thinking embedded in play

The Three Projects

TIMER · GRAMOPHONE · FORTUNE WHEEL

Timer

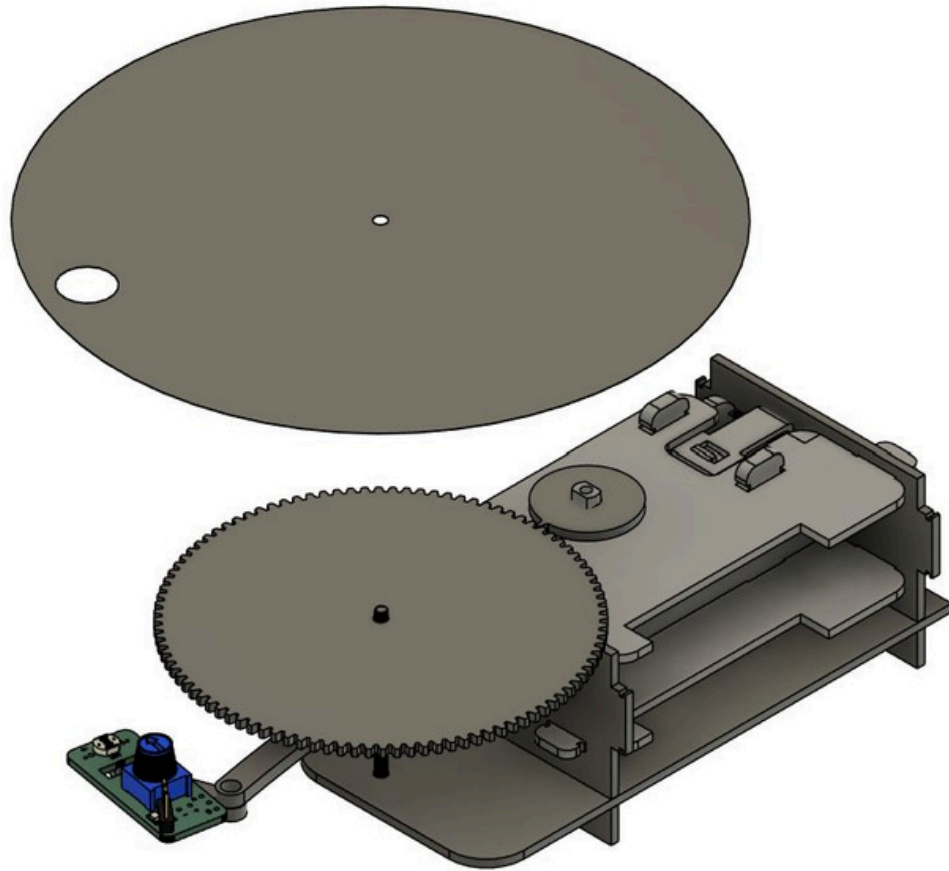
100 RPM motor with 100:1 gear reduction. Disc slit aligns with LDR after exactly 1 minute → buzzer fires. **No microcontroller.**

Gramophone

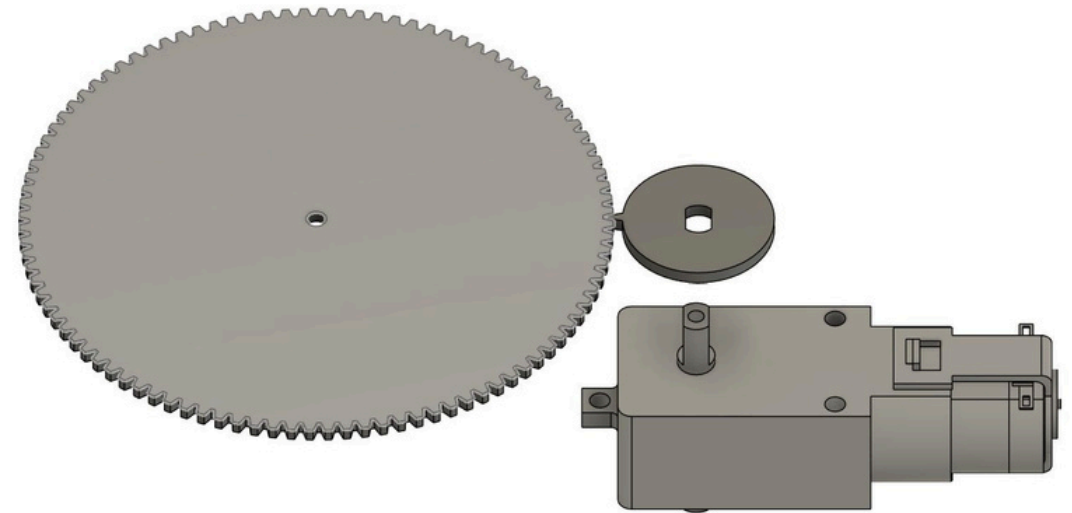
Calculated slit spacings on a rotating disc. Each slit triggers a light pulse → buzzer beep. Slit positions control the **rhythm and melody.**

Fortune Wheel

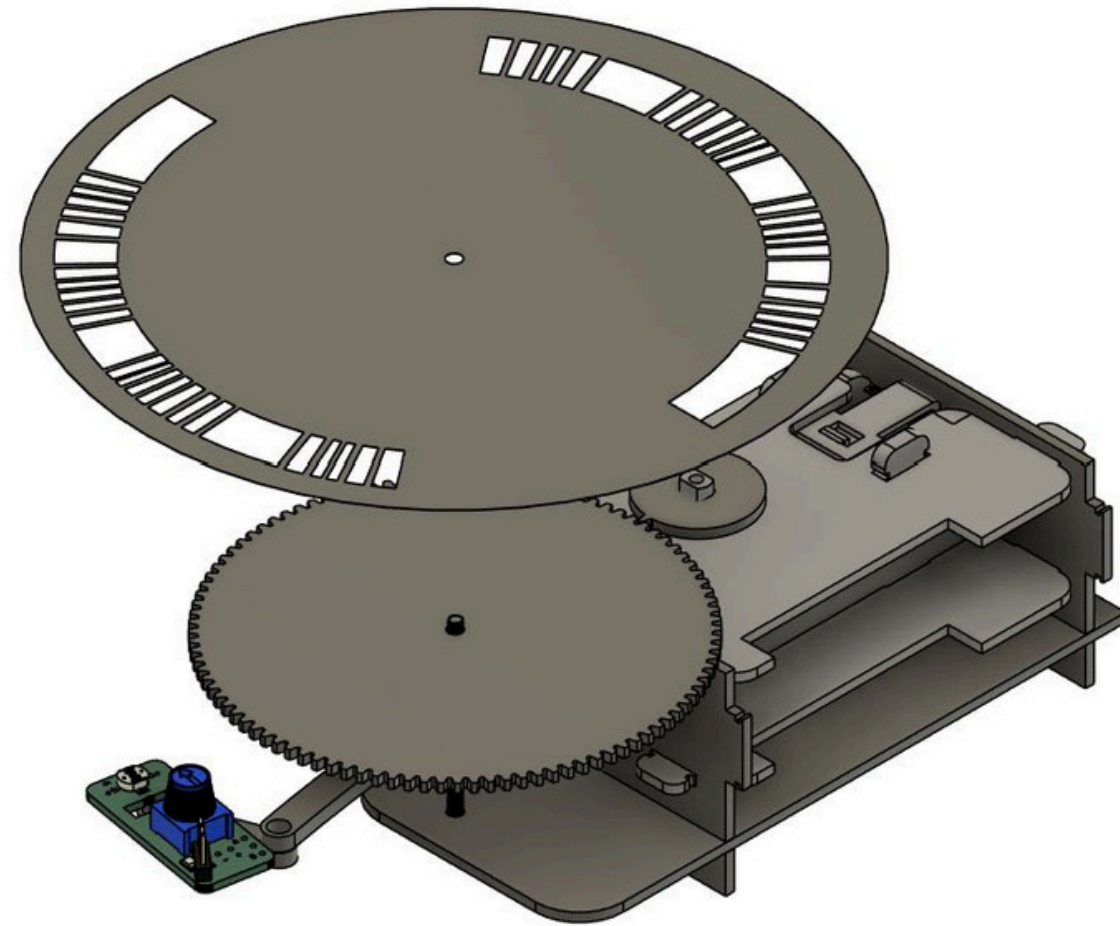
Childspins disc manually. Slit over LDR → motor on; closed section → motor off. **Uncertain by design** — teaches probability and input-output logic.

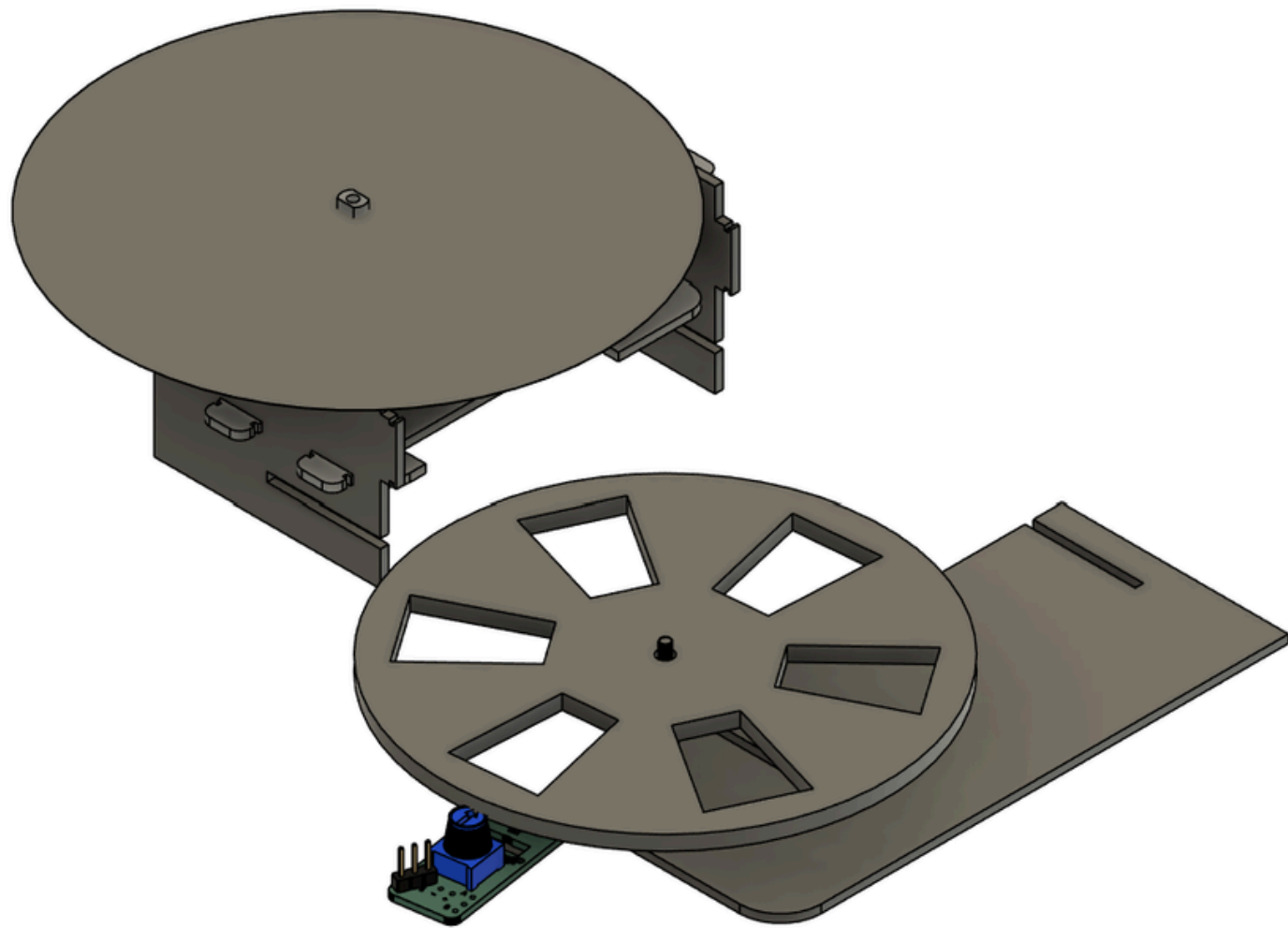
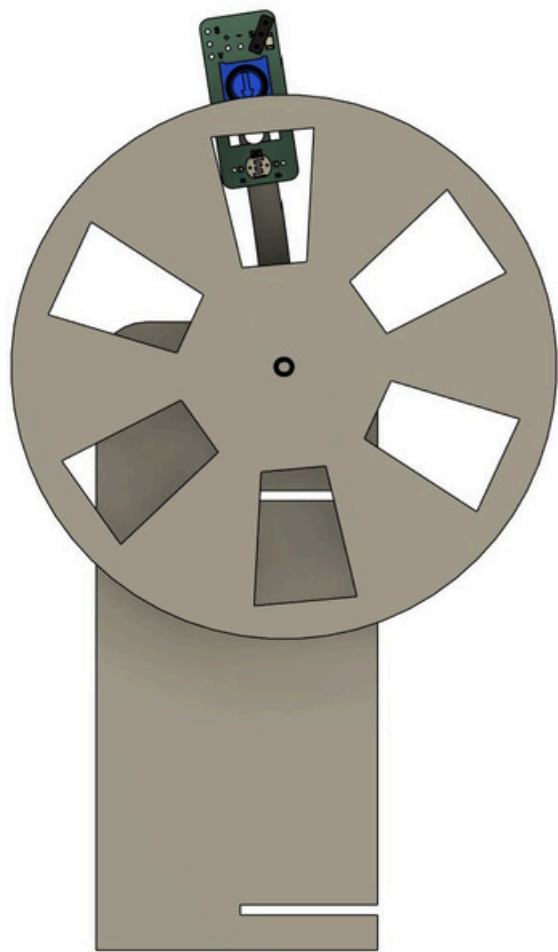


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Key Learnings

WHAT 5 MONTHS ACTUALLY TAUGHT ME



Designing for Children

Psychology and ergonomics must lead every decision.



Electronics Integration

PCBs, LDRs, motor drivers, and LED systems built from scratch.



DFM Thinking

Creative vision must survive BOM constraints and cost reality.



Iterative Prototyping

stages from concept to production is normal — not failure.



Visual Instruction Design

Complexity compressed into illustrations a child follows alone.



Real Testing = Design

Workshops are not validation—they are design tools.

Thank You

Design is not about products. It's about what those products do to people.

OiMake

For the opportunity, trust, and resources.

Prof. Bhushan Sonawane

For continuous guidance and mentorship.

Woxsen University, SOAD

For the foundation.

The Jury

For your time and feedback.